CURRICULUM VITAE

Shampa Banik

Oulu, Finland

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RESEARCH INTERESTS

Artificial Intelligence, Machine Learning, Image Processing, Computer Vision, NLP

EDUCATION

University of Oulu, Finland August 2026 — Present
M.Sc. in Technology, Computer Science and Engineering (Artificial Intelligence Track)

Patuakhali Science & Technology University (PSTU) January 2018 — October 2023
B.Sc. in Computer Science and Engineering (Dean's List for the last three years) Cumulative GPA: 3.84/4.00 (Rank: 1/46)
Thesis Title: Prediction of Dengue Outbreaks Using LSTM Recurrent Neural Network in Bangladesh

APPOINTMENTS

Bangladesh University of Business and Technology (BUBT) Dhaka, Bangladesh
Lecturer July 2024 — Present

Conducting Courses: System Analysis and Design, Numerical Analysis, Discrete Mathematics, Computer Networks and Computer Operation & Applications.
Responsibilities: Preparing lectures, teaching, invigilating exams, conducting research, student evaluation, academic guidance, supervision, intake management, and participating in committee and departmental work

University of Scholars Dhaka, Bangladesh
Lecturer November 2023 — June 2024

Conducting courses: Data Structure and Algorithms, Software Engineering, Computer Fundamentals, Numerical Analysis, Competitive Programming, and Technical Writing.
Responsibilities: Preparing lectures, teaching, invigilating exams, evaluating students, conducting research, providing academic guidance, supervising students, and participating in departmental work

HONORS AND AWARDS

PSTU Dean's Merit Award: Awarded scholarship for academic excellence among 46 students from session 2019 to 2021.
PSTU IT Carnival: First Prize for the "IOT-Based Food Spoilage Detection System" project at IT Carnival.
NGPC: Competed in the final round of **National Girls Programming Contest (NGPC)** in both 2019 and 2021.
PSTU Independence Day Programming Contest: Ranked in the top 5 out of 50 teams in 2018.
Merit Scholarships: Government Board Merit scholarships for outstanding results in both Secondary School Certificate (2015) and Junior School Certificate (2012).

PUBLICATIONS

Conference Papers

1. **Banik, S.,** Shahid, S. B., & Afroj, M. (2025, July). ColoNetBlock: Attention enhanced CNN for colorectal cancer classification with visual explanation. In Proceedings of the 2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS) (pp. 1–6). IEEE. [Paper]
2. Sharma, R., **Banik, S.,** Hossain, M. Z., Kabir, M. F., Thapa, S., Ullah, M. A., ... & Bhavani, G. D. (2024, December). A Novel Classification Scheme for Early Alzheimer's Disease (AD) Severity Diagnosis Using Deep Features of a Hybrid Cascade Attention Architecture. In 2024 International Conference on Augmented Reality, Intelligent Systems, and Industrial Automation (ARIIA) (pp. 1-6). IEEE. [Paper]
3. Arman, M., **Banik, S.**, Rusho, M. A., Sayem, S., Rahman, M. M., Kabir, T., ... & Jahan, M. K. (2025, April). Hyperparameter Optimization: Achieving State-of-the-Art Caloric Burn Prediction with Fine-Tuned MLPRegressor. In 2025 IEEE 4th International Conference on Computing and Machine Intelligence (ICMI) (pp. 1-6). IEEE. [Paper]
4. Khan, A., Valivarthi, D. T., **Banik, S.,** Kakkad, A., Sujatha, B., & Acharya, D. (2025, February). Development of Additive Manufacturing Design Based on Machine Learning: A Biomedical Engineering Approach. In 2025 First International Conference on Advances in Computer Science, Electrical, Electronics, and Communication Technologies (CE2CT) (pp. 817-821). IEEE [Paper]

5. Porna, S. B., Rahman, M. A., **Banik, S.,** Khan, M. H., Uddin, R., Basit, A., ... & Bhavani, G. D. (2025, April). Novel CNN-Based Models for Breast Ultrasound Image Classification: Leveraging EEHO, PSO, and BSA Optimizers. In 2025 International Conference on Knowledge Engineering and Communication Systems (ICKECS) (pp. 1-6). IEEE.[Paper]
6. Hossain, T., Pramanik, A., **Banik, S.,** & Islam, J. (2025, July). Classifying Bengali Quotes: A Comparative Study of Machine Learning and Deep Learning Approaches. In 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN) (pp. 1-5). IEEE.[Paper]
7. Joy, M. A. M., Sharmin, S., **Banik, S.,** & Arif, A. (2026, January). A Low-Resource Transformer-Based Deep Learning Framework for Precise Localization and Classification of Visible Crop Diseases. In 2026 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE) (pp. 1-6). IEEE.[Paper]

SELECTED PROJECTS

D-Alert: Dengue Prediction using Machine Learning | Python *Jan 2023*

- Built a dengue dataset from DGHS data, developed an LSTM model for area-based dengue prediction, and benchmarked against multiple baseline models.

Eatsy: Web Application for Food Ordering | HTML, CSS, Bootstrap, PHP, JavaScript, SQL *August 2022*

- Developed a smart food ordering app using modern development tools and software design principles to ensure fast connectivity, efficient data management, and a modular system architecture.

WellnessWise: A Desktop Application for Healthcare Management | Java, System Design *August 2022*

- Built a healthcare management desktop app using Java and SQL, employing OOP principles for efficient management of patient records and appointments, ensuring modularity and scalability.

SmartSpoil: IOT Based Food Spoilage Detection System | Arduino *July 2021*

- Designed and implemented an IoT-based food spoilage detection system using Arduino and sensor data, providing real-time freshness alerts; awarded First Prize at PSTU IT Carnival. The system utilizes sensors to track key freshness indicators, providing real-time alerts to ensure food safety.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, C++, Javascript, MATLAB, HTML, CSS, Assembly
- **Machine Learning:** PyTorch, Keras, OpenCV, TensorFlow, NumPy, NLTK, Pandas, Matplotlib
- **Developer Tools:** VS Code, CodeBlocks, LaTeX, PyCharm, Jupyter Notebook, Eclipse
- **Frameworks:** Bootstrap, Arduino
- **Statistics & Analytics:** SPSS
- **Version Control:** Git, Github
- **Illustration Tools:** Proficient in Adobe Illustrator, Adobe Photoshop

SELECTED CO-CURRICULAR ACTIVITIES AND SERVICES

- Created hundreds of **digital artworks and logo designs**
- **Member, Sub-Committee (Project), CSE Club, PSTU**
  Contributed to organizing and implementing projects, fostering innovation within the club. September 2021 - October 2023
- **Member, Debating Society, PSTU**
  Contributed to organizing events, preparing debate topics, mentoring new members, and representing the club in competitions. *January 2019 - October 2023*
- **Member, BADHAN, Blood Donation Organization**
  Participated in organizing and promoting blood donation drives, contributing to raising awareness and encouraging community involvement. *December 2018 - January 2023*
- **Volunteer, Communication Team – Talk Digital Platform (Way to Dreams):**
  Assisted with communication efforts, audience engagement, and content outreach. August 2021 - October 2021
- An Article published on Medium in 2024 - **Beyond the Boom: Unraveling the Future of Explainable AI in 2030**

WORKSHOPS

- **International Workshop on Wireless Networking and Cyber Security (WWNCS)**
- **Regional Workshop on Networking Basics and Campus Network Design (WNBCND-1)**
- **Workshop on Skill Development for Mobile Game and Application Project (Android)**

TRAINING AND ONLINE COURSES

- **CCNA: Introduction to Networks, Switching, Routing, and Wireless Essentials, Enterprise Networking, Security, and Automation**, Issued by Cisco.
- **Machine Learning Specialization**, University of Washington, Coursera.